

Project Title

Design and Implementation of a Smart Secure Enterprise Campus Network with IoT-Based Monitoring and High Availability

Abstract

Modern organizations require reliable, scalable, secure and efficient network infrastructures to support communication, resource sharing and service delivery. This project presents the design and implementation of a Smart Multi-Campus Enterprise Network using Cisco Packet Tracer.

The network consists of two campuses: Campus A (Main Campus) and Campus B (Branch Campus). The infrastructure follows a hierarchical Core-Access architecture and incorporates multiple advanced networking technologies including VLAN segmentation, Inter-VLAN Routing, Dynamic Routing using OSPF, HSRP for gateway redundancy, EtherChannel for link aggregation, DHCP for automatic IP assignment, DNS, HTTP, Email services, ACL-based security, NAT, Wireless networking and IoT integration.

The implementation provides secure communication between departments while restricting unauthorized access through Access Control Lists. Network availability is improved through HSRP and EtherChannel. Wireless connectivity is provided using WPA2-PSK secured access points. IoT devices including Motion Detector, Smoke Detector, Siren and Webcam are integrated to simulate a smart campus environment.

The project demonstrates how enterprise-level networking technologies can be combined to create a scalable, manageable and highly available network infrastructure.

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Chapter 1: Introduction

Computer networking plays a vital role in modern organizations. Educational institutions, enterprises and industries rely on computer networks for communication, data sharing and centralized resource management.

As organizations grow, network infrastructures become more complex. A single flat network becomes inefficient due to increased broadcast traffic, security concerns and management difficulties. Therefore, modern networks use segmentation, routing and security technologies to improve performance and reliability.

This project implements a Smart Multi-Campus Enterprise Network using Cisco Packet Tracer. The network is designed for Pundra University and consists of a Main Campus (Campus A) and a Branch Campus (Campus B). Various networking technologies have been implemented to ensure scalability, security, availability and efficient resource utilization.

The network includes departmental VLANs, dynamic routing, server services, wireless connectivity and IoT-based monitoring systems. The project demonstrates real-world enterprise networking concepts and practical implementation techniques.

Chapter 2: Project Objectives

The major objectives of this project are:

1. Design a scalable multi-campus network architecture.
2. Separate departments using VLAN technology.
3. Reduce broadcast traffic through subnetting.
4. Enable communication between VLANs through Layer-3 switching.
5. Implement dynamic routing using OSPF.

6. Configure DHCP for automatic IP assignment.
7. Implement DNS service for domain name resolution.
8. Configure HTTP and HTTPS services.
9. Configure Email services using SMTP and POP3.
10. Provide network redundancy using HSRP.
11. Increase bandwidth and redundancy using EtherChannel.
12. Implement security using ACL and Port Security.
13. Configure NAT for address translation.
14. Provide secure wireless connectivity.
15. Integrate IoT devices into the network.
16. Simulate a real-world enterprise environment.

Chapter 3: Network Design Overview

The network consists of two campuses connected through a routed WAN connection.

Campus A (Main Campus)

Campus A contains the primary enterprise infrastructure including:

- Core Switch 1 (CORE-SW1)
- Core Switch 2 (CORE-SW2)
- Multiple Access Switches
- Centralized Server Room
- Wireless Network
- IoT Devices

Departments include:

- Admin
- Faculty
- Student
- Library
- Accounts
- IoT
- Server
- Guest
- Management

Campus B (Branch Campus)

Campus B contains:

- CORE-SW3
- Access Switches
- Branch Users

Departments include:

- B-Admin
- B-Student
- B-Faculty
- B-Lab
- B-Management

Core-Access Architecture

The project follows a hierarchical design model.

Core Layer

The Core Layer is responsible for:

- High-speed switching
- Routing
- Redundancy
- Inter-campus connectivity

Devices:

- CORE-SW1
- CORE-SW2
- CORE-SW3

Access Layer

The Access Layer provides connectivity to end-user devices.

Functions:

- VLAN assignment
- Port Security
- End-device connectivity

Devices:

- Access Switches
- Wireless Access Points

Network Technologies Used

- VLAN
- Subnetting
- Inter-VLAN Routing
- OSPF
- HSRP
- EtherChannel
- DHCP

- DNS
- HTTP
- HTTPS
- Email Server
- ACL
- Port Security
- NAT
- Wireless Security
- IoT Integration

These technologies collectively provide a secure, scalable, reliable and manageable enterprise network.

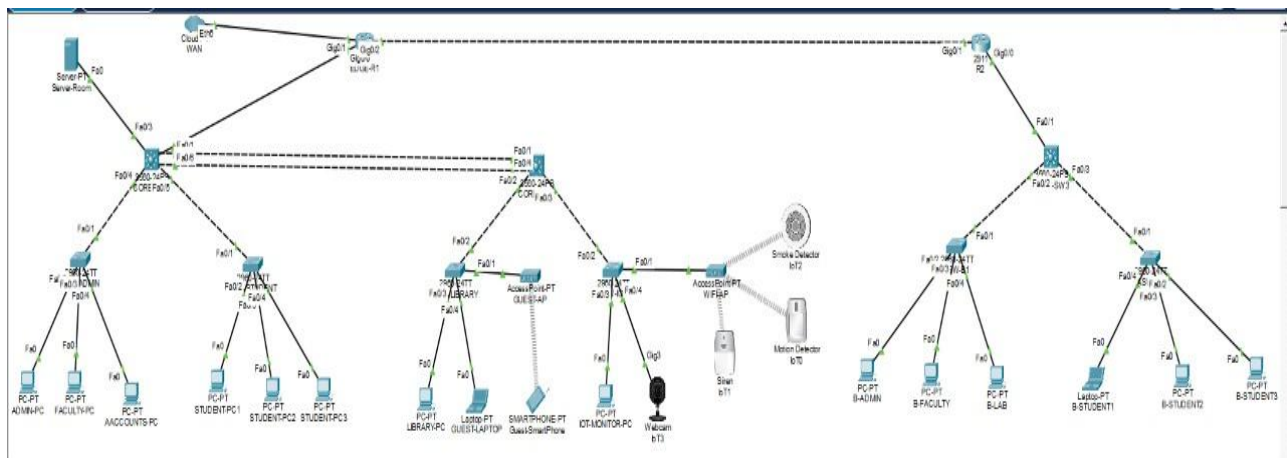


Figure 3.1: Complete Multi-Campus Enterprise Network Topology

Chapter 4: VLAN Design and Subnetting

VLAN Implementation

Network segmentation was implemented using Virtual Local Area Networks (VLANs). Each department was assigned a dedicated VLAN to improve security, reduce broadcast traffic and simplify network management.

Campus A VLAN Table

VLAN ID	Department	Network
10	Admin	192.168.10.0/24
20	Faculty	192.168.20.0/24
30	Student	192.168.30.0/24
40	Library	192.168.40.0/24
50	Accounts	192.168.50.0/24
60	IoT	192.168.60.0/24
70	Server	192.168.70.0/24
80	Guest	192.168.80.0/24
99	Management	192.168.99.0/24

Campus B VLAN Table

VLAN ID	Department	Network
110	B-Admin	10.10.10.0/24
120	B-Student	10.10.20.0/24
130	B-Faculty	10.10.30.0/24
140	B-Lab	10.10.40.0/24
999	B-Management	10.10.99.0/24

What is Subnetting?

Subnetting is the process of dividing a large network into multiple smaller logical networks.

Why Subnetting Was Used

1. Reduce broadcast traffic.
2. Improve security.
3. Simplify management.
4. Improve network performance.
5. Separate departments logically.

Problems Without Subnetting

- Excessive broadcast traffic.
- Poor performance.
- Difficult management.
- Security risks.
- No departmental isolation.

VLSM Discussion

Variable Length Subnet Masking (VLSM) allows different subnet sizes according to host requirements.

Why VLSM is Better

- Efficient IP utilization.
- Reduced address wastage.
- Better scalability.
- Flexible network design.

Advantages of VLSM

- Efficient use of address space.
- Better network planning.
- Supports different department sizes.

Disadvantages of VLSM

- More complex configuration.
- More difficult troubleshooting.
- Requires careful planning.

Why VLSM Was Not Implemented

The current design uses fixed-length subnetting (/24) to maintain simplicity, stability and easier management. VLSM can be considered as a future optimization.

Chapter 5: IP Address Planning

Campus A Gateway Plan

VLAN	SVI Address
VLAN10	192.168.10.1
VLAN20	192.168.20.1
VLAN30	192.168.30.1
VLAN40	192.168.40.1
VLAN50	192.168.50.1
VLAN60	192.168.60.1
VLAN70	192.168.70.1
VLAN80	192.168.80.1
VLAN99	192.168.99.1

Campus B Gateway Plan

VLAN	SVI Address
VLAN110	10.10.10.1
VLAN120	10.10.20.1
VLAN130	10.10.30.1
VLAN140	10.10.40.1
VLAN999	10.10.99.2

Server Information

Device	IP Address
Server-Room	192.168.70.2

WAN Addressing

Device	Interface	Address
EDGE-R1	G0/2	172.16.1.1/30
R2	G0/1	172.16.1.2/30

Chapter 6: Inter-VLAN Routing

Inter-VLAN Routing was implemented using Layer-3 Switching.

The multilayer core switches provide routing services between VLANs.

Why Inter-VLAN Routing Is Required

Different VLANs are separate broadcast domains and cannot communicate directly.

Inter-VLAN routing allows:

- Admin↔ Server communication.
- Faculty↔ Server communication.
- Campus communication.
- Centralized resource access.

Limitation Without Inter-VLAN Routing

Without Inter-VLAN Routing:

- VLANs remain isolated.
- Server access becomes impossible.
- Resource sharing cannot occur

Chapter 7: Dynamic Routing Using OSPF

Open Shortest Path First (OSPF) was implemented as the dynamic routing protocol.

OSPF Configuration

Process ID:

OSPF Process 1

Area:

Area 0

WANNetwork

172.16.1.0/30

Benefits of OSPF

1. Fast convergence.
2. Better scalability.
3. Supports large enterprise networks.
4. Efficient route calculation.
5. Supports hierarchical design.

Why Dynamic Routing Was Used

Dynamic routing automatically updates routes when network changes occur.

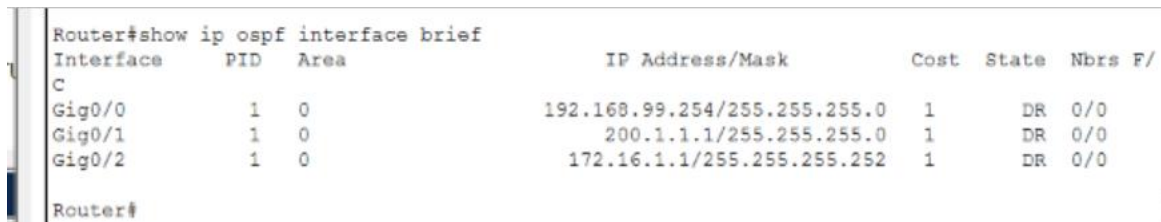
Static Routing Limitations

- Manual configuration.
- Difficult maintenance.
- Poor scalability.

OSPF vs RIP

Feature	OSPF	RIP
Metric	Cost	Hop Count
Convergence	Fast	Slow
Scalability	High	Low
Enterprise Use	Yes	Limited

Verification



```
Router#show ip ospf interface brief
Interface      PID  Area          IP Address/Mask      Cost  State  Nbrs F/
C
Gig0/0         1    0             192.168.99.254/255.255.255.0  1     DR    0/0
Gig0/1         1    0             200.1.1.1/255.255.255.0    1     DR    0/0
Gig0/2         1    0             172.16.1.1/255.255.255.252  1     DR    0/0
Router#
```

Figure 7.1: OSPF Routing Table (Campus A)

Chapter 8: HSRP Configuration

Hot Standby Router Protocol (HSRP) was implemented on Campus A core switches.

Purpose

Provide gateway redundancy and high availability.

HSRP Virtual Gateways

VLAN	Virtual Gateway
VLAN10	192.168.10.254
VLAN20	192.168.20.254
VLAN30	192.168.30.254
VLAN40	192.168.40.254
VLAN50	192.168.50.254
VLAN60	192.168.60.254

VLAN	Virtual Gateway
VLAN70	192.168.70.254
VLAN80	192.168.80.254

Active Device

CORE-SW1

Priority: 120

Standby Device

CORE-SW2

Priority: 100

Benefits

- Gateway redundancy.
- Automatic failover.
- High availability.
- Reduced downtime.

Verification

```
Switch#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State   Active        Standby        Virtual IP
Vl10       1    120  P Active  local         192.168.10.2   192.168.10.254
Vl20       2    120  P Active  local         192.168.20.2   192.168.20.254
Vl30       3    120  P Active  local         192.168.30.2   192.168.30.254
Vl40       4    120  P Active  local         192.168.40.2   192.168.40.254
Vl50       5    120  P Active  local         192.168.50.2   192.168.50.254
Vl60       6    120  P Active  local         192.168.60.2   192.168.60.254
Vl70       7    120  P Active  local         192.168.70.3   192.168.70.254
Vl80       8    120  P Active  local         192.168.80.2   192.168.80.254
```

Figure 8.1: HSRP Standby Gateway

Chapter 9: EtherChannel Configuration

EtherChannel was configured between core switches using LACP.

Protocol

LACP (Link Aggregation Control Protocol)

Port-Channel

Port-Channel 1

CORE-SW1 Members

- Fa0/1
- Fa0/6

CORE-SW2 Members

- Fa0/1
- Fa0/4

Benefits

1. Increased bandwidth.
2. Load balancing.
3. Redundancy.
4. Fault tolerance.
5. Reduced congestion.

Verification

Command:

```
Switch#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)          LACP        Fa0/1 (P) Fa0/6 (P)
Switch#
```

Figure 9.1: EtherChannel Configuration

Chapter 10: DHCP Configuration

Dynamic Host Configuration Protocol (DHCP) was implemented using the centralized server.

Server IP:

192.168.70.2

Campus A DHCP Pools

- ADMIN
- FACULTY
- STUDENT
- LIBRARY
- ACCOUNTS
- IOT
- GUEST

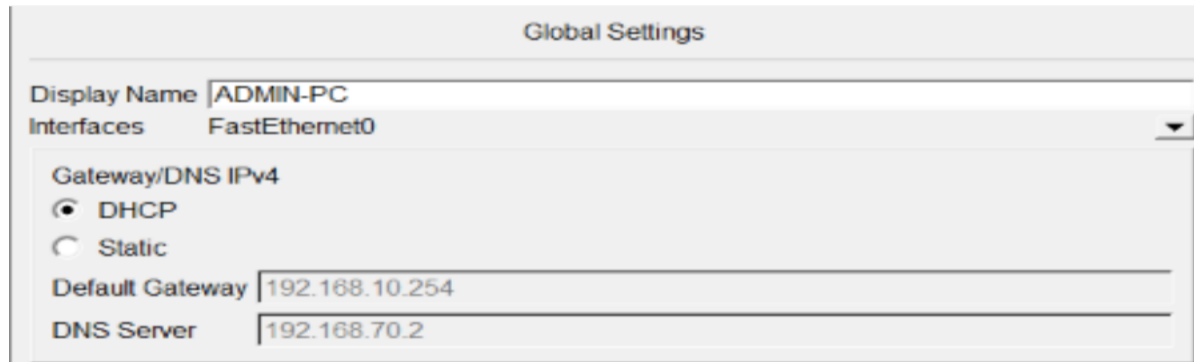
Campus B DHCP Pools

- ADMIN-B
- STUDENT-B
- FACULTY-B
- LAB-B

Why DHCP Was Used

1. Automatic IP assignment.
2. Easier management.
3. Reduced configuration errors.
4. Faster deployment.

DHCP Verification



The image shows a 'Global Settings' window with the following configuration:

Global Settings	
Display Name	ADMIN-PC
Interfaces	FastEthernet0
Gateway/DNS IPv4	
☒ DHCP	
☐ Static	
Default Gateway	192.168.10.254
DNS Server	192.168.70.2

Figure 10.1: Automatic IP Assignment Using DHCP

Result:

All client devices successfully received IP addresses automatically.

Chapter 11: DNS Configuration

Domain Name System (DNS)

DNS was configured on the centralized server to provide hostname-to-IP address resolution.

Server Information

Server IP Address:

192.168.70.2

DNS Service:

Enabled

DNS Records

Domain Name	IP Address
www.PUC.local	192.168.70.2
mail.PUC.com	192.168.70.2

Why DNS Was Used

DNS allows users to access services using domain names instead of remembering IP addresses.

Advantages of DNS

1. Easy access to resources.
2. User-friendly service access.
3. Centralized name management.
4. Improved usability.

Verification

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.puc.local

Pinging 192.168.70.2 with 32 bytes of data:

Reply from 192.168.70.2: bytes=32 time=1ms TTL=127
Reply from 192.168.70.2: bytes=32 time<1ms TTL=127
Reply from 192.168.70.2: bytes=32 time=14ms TTL=127
Reply from 192.168.70.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.70.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 3ms
```

Figure 11.1:Successful pinged

Chapter 12: HTTP and HTTPS

Configuration Web Server Configuration

The server was configured to provide web services using HTTP and HTTPS protocols.

Server IP

192.168.70.2

Services Enabled

- HTTP
- HTTPS

Hosted Files

- index.html
- index.php

Purpose

The web server provides centralized access to enterprise web resources.

Verification

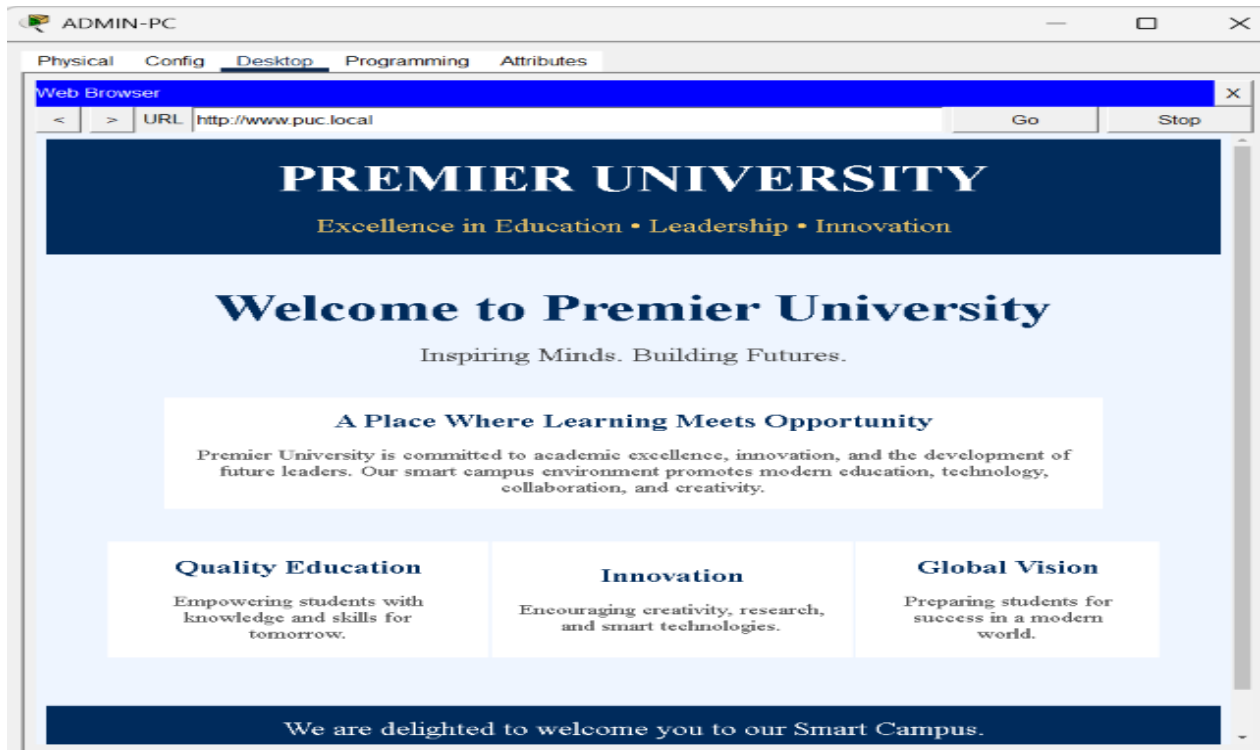


Figure 12.1: Successful Website Access

Benefits

1. Internal web hosting.
2. Centralized information sharing.
3. Browser-based service access.

Chapter 13: Email Server Configuration

Email Service

Email communication was implemented using SMTP and POP3 protocols.

Email Domain

PUC.com

Protocols Used

SMTP (Simple Mail Transfer Protocol)

POP3 (Post Office Protocol Version 3)

Configured Users

Campus A:

- admin
- faculty
- student
- accounts
- library
- iot

Campus B:

- Badmin
- Bfaculty
- Bstudent
- Blab

Example Email Addresses

admin@PUC.com

student@PUC.com

faculty@PUC.com

accounts@PUC.com

Purpose

Email services provide communication between users across departments and campuses.

Verification

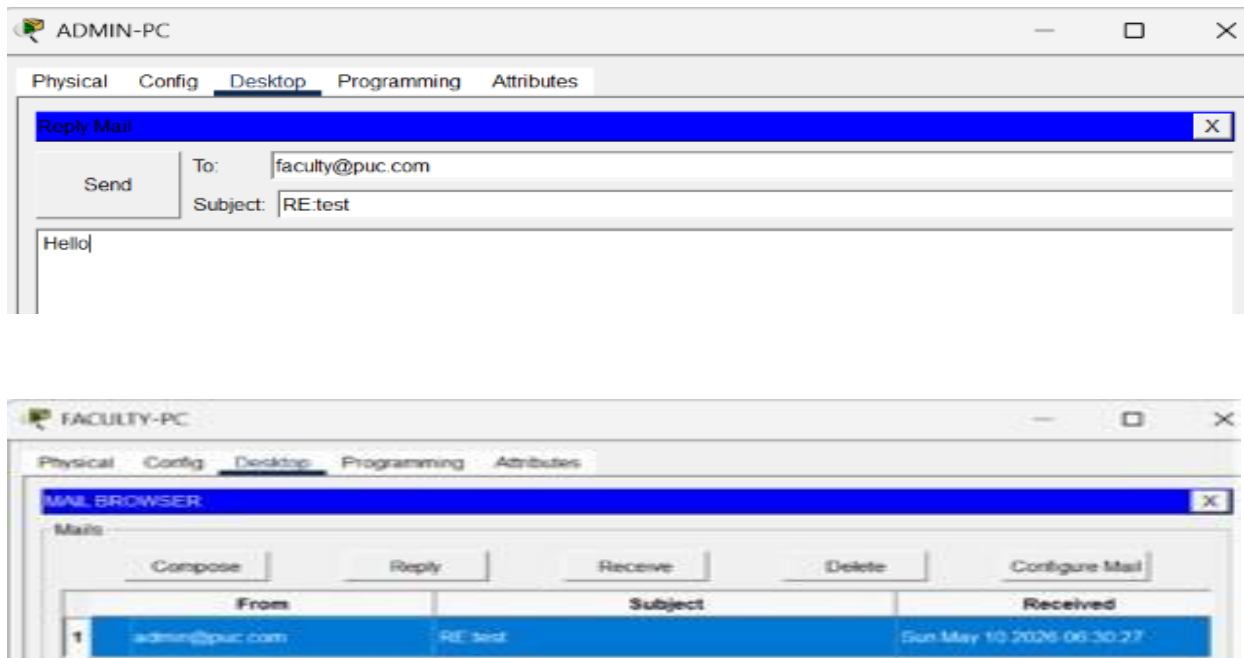


Figure 13.1: Email Transmission Test

Benefits

1. Internal communication.
2. Professional enterprise service.
3. Centralized messaging platform.

Chapter 14: Access Control List (ACL)

ACL Implementation

Access Control Lists were implemented to restrict unauthorized communication between departments.

Security Policy

Student users are restricted from accessing critical enterprise resources.

Campus A ACL Rules

Student VLAN:

192.168.30.0/24

Blocked Access:

- Admin VLAN (192.168.10.0/24)
- Accounts VLAN (192.168.50.0/24)
- Server VLAN (192.168.70.0/24)
- Campus B Admin Network

Campus B ACL Rules

Student VLAN:

10.10.20.0/24

Blocked Access:

- Campus B Admin
- Campus A Admin
- Campus A Accounts
- Campus A Server

Purpose

1. Unauthorized access prevention.
2. Department isolation.
3. Enterprise security enforcement.
4. Protection of critical resources.

Verification

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.70.2

Pinging 192.168.70.2 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.70.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 14.1: ACL Access Restriction Verification

Chapter 15: Port Security

Port Security Configuration

Port Security was implemented on access switches to prevent unauthorized devices from connecting to the network.

Configuration

Maximum MAC Address:

1

Violation Mode:

Shutdown

Protected Interfaces

Campus A:

- Fa0/2
- Fa0/4

Campus B:

- Fa0/2
- Fa0/3

Purpose

1. Prevent unauthorized access.
2. Restrict rogue devices.
3. Improve physical security.
4. Protect access layer infrastructure.

Verification

```
Switch#show port-security
```

Secure	Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security	Action
	Fa0/3	1	1	0		Shutdown

```
Switch#
```

Figure 15.1: Port Security Verification

Chapter 16: NAT Configuration

Network Address Translation

NAT was implemented on EDGE-R1 to enable communication between private and public networks.

NAT Type

PAT (Port Address Translation)

Inside Interface

GigabitEthernet0/0

IP:

192.168.99.254

Outside Interface

GigabitEthernet0/1

IP:

200.1.1.1

NAT ACL

access-list 1 permit 192.168.0.0 0.0.255.255

NAT Rule

ip nat inside source list 1 interface GigabitEthernet0/1 overload

Benefits

1. Public IP conservation.
2. Improved security.
3. Address translation.
4. Internet access simulation.

Verification

```
Router#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 200.1.1.1:21      192.168.30.12:21 200.1.1.2:21      200.1.1.2:21
icmp 200.1.1.1:22      192.168.30.12:22 200.1.1.2:22      200.1.1.2:22
icmp 200.1.1.1:23      192.168.30.12:23 200.1.1.2:23      200.1.1.2:23
icmp 200.1.1.1:24      192.168.30.12:24 200.1.1.2:24      200.1.1.2:24
```

Figure 16.1: NAT Translation Table Verification

Chapter 17: Wireless Network Configuration

Wireless Infrastructure

A secure wireless network was implemented for guest users.

SSID: PUC_Guest

Authentication: WPA2-PSK

Encryption: AES

Password: guest12345

Wireless Client

Guest-SmartPhone

Assigned IP:

192.168.80.10

IP Assignment Method

DHCP

Purpose

1. Mobility.
2. Guest access.
3. Secure wireless communication.

Verification

Wireless client successfully connected and received an IP address.

Chapter 18: IoT Integration

Smart Campus Monitoring System

IoT devices were integrated into the network to create a smart enterprise environment.

Devices Used

1. Motion Detector
2. Smoke Detector
3. Siren
4. Webcam

Monitoring Device

IOT-MONITOR-PC

Benefits

1. Smart security.
2. Real-time monitoring.
3. Automated alerts.
4. Improved safety.

Chapter 19: Discussion

The project successfully implemented a secure and scalable multi-campus enterprise network infrastructure using Cisco Packet Tracer.

The integration of VLANs, OSPF, HSRP, EtherChannel, DHCP, DNS, Email services, ACLs, NAT, Wireless networking and IoT technologies demonstrates a realistic enterprise networking environment.

The network achieved departmental isolation, automatic IP management, secure communication and high availability. Redundancy mechanisms such as HSRP and EtherChannel improved reliability while ACLs and Port Security enhanced security.

The project successfully met all mandatory requirements and incorporated additional bonus features including NAT, HSRP, EtherChannel and IoT integration.

Limitations

1. Implemented in simulation environment.
2. VLSM was not fully implemented.
3. No real internet connectivity.
4. Limited advanced monitoring tools.
5. IPv6 was not implemented.

Future Improvements

1. Full VLSM implementation.
2. IPv6 deployment.
3. Advanced firewall integration.
4. Network monitoring system implementation.
5. Cloud service integration.
6. VPN deployment.
7. IDS/IPS implementation.
8. Wireless controller deployment.
9. Multi-area OSPF implementation.
10. Enterprise-grade redundancy expansion.